

VILYUYSK, Russian Federation

WMO: 246410

Lat: 63.783N

Lon: 121.617E

Elev: 111

StdP: 100.00

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.8	-46.7	-51.0	0.0	-48.7	-48.9	0.0	-46.7	5.0	-26.7	4.3	-26.1	0.8	240	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	30.4	19.1	28.3	18.0	26.1	17.0	20.0	28.4	18.8	26.6	17.7	24.7	2.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.1	23.3	15.7	11.3	22.6	14.6	10.5	21.9	57.8	28.4	54.1	26.7	50.6	24.9	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.5	4.0	DB	-48.8	33.6	2.4	1.9	-50.5	35.0	-51.9	36.2	-53.2	37.2	-55.0	38.6	(4)
(5)			WB	-48.8	21.6	2.4	1.1	-50.5	22.4	-51.9	23.0	-53.2	23.6	-55.0	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.7	-35.0	-30.4	-18.1	-4.9	6.1	16.1	18.8	14.7	5.9	-7.0	-24.9	-34.8	(6)
(7)		DBStd	20.49	7.58	7.38	7.90	5.98	4.98	5.00	4.14	3.89	4.59	6.78	8.47	7.74	(7)
(8)		HDD10.0	7102	1395	1131	871	447	143	9	1	7	139	526	1046	1388	(8)
(9)		HDD18.3	9593	1653	1364	1129	697	379	100	46	124	374	784	1296	1646	(9)
(10)		CDD10.0	654	0	0	0	0	22	191	274	153	14	0	0	0	(10)
(11)		CDD18.3	102	0	0	0	0	0	32	60	11	0	0	0	0	(11)
(12)		CDH23.3	1154	0	0	0	0	6	390	641	117	1	0	0	0	(12)
(13)		CDH26.7	369	0	0	0	0	1	123	223	22	0	0	0	0	(13)
(14)	Wind	WSAvg	1.9	1.5	1.5	1.6	2.0	2.3	2.2	2.0	1.9	2.0	2.0	1.7	1.5	(14)
(15)	Precipitation	PrecAvg	266	11	8	9	11	22	34	48	30	24	19	14	(15)	
(16)		PrecMax	416	27	32	23	31	64	70	128	140	95	48	39	30	(16)
(17)		PrecMin	143	2	0	1	0	0	10	0	6	6	5	5	3	(17)
(18)		PrecStd	55	6	5	6	8	15	17	29	27	17	10	8	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.5	-7.1	4.6	12.4	24.1	32.5	34.0	29.1	22.0	6.5	-0.7	-10.8	(19)
(20)			MCWB	-11.1	-8.1	0.6	6.0	13.4	20.0	21.4	18.9	13.8	3.6	-2.0	-11.4	(20)
(21)		2%	DB	-16.5	-11.9	0.2	8.8	20.8	30.0	31.3	26.5	18.5	4.4	-4.8	-16.8	(21)
(22)			MCWB	-16.9	-12.6	-2.7	3.7	11.4	18.4	19.9	17.3	11.5	2.2	-5.8	-17.2	(22)
(23)		5%	DB	-20.5	-16.3	-2.7	6.4	17.8	27.8	29.2	24.2	15.6	2.6	-8.6	-20.3	(23)
(24)			MCWB	-20.9	-16.8	-4.9	1.8	9.7	17.2	18.8	16.1	10.2	0.9	-9.2	-20.6	(24)
(25)		10%	DB	-24.1	-19.6	-5.4	4.4	15.2	25.3	27.1	22.2	13.1	1.1	-12.2	-23.2	(25)
(26)			MCWB	-24.3	-20.0	-7.1	0.5	8.2	15.9	17.7	15.1	8.8	-0.3	-12.7	-23.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.1	-8.0	0.8	6.1	13.7	20.6	22.2	19.7	14.4	4.4	-1.8	-11.2	(27)
(28)			MCDWB	-10.4	-7.3	4.5	11.8	23.1	30.9	32.2	26.5	20.8	6.0	-0.7	-10.6	(28)
(29)		2%	WB	-16.9	-12.5	-2.5	3.9	11.6	19.2	20.6	18.2	12.3	2.5	-5.8	-17.2	(29)
(30)			MCDWB	-16.5	-11.9	0.3	8.5	20.0	28.6	29.2	24.7	16.6	3.9	-5.0	-16.9	(30)
(31)		5%	WB	-20.8	-16.8	-4.8	2.1	10.0	17.9	19.6	17.0	10.7	1.0	-9.2	-20.6	(31)
(32)			MCDWB	-20.5	-16.4	-2.7	6.0	17.2	26.4	27.4	22.5	15.0	2.4	-8.6	-20.3	(32)
(33)		10%	WB	-24.3	-20.0	-7.1	0.7	8.5	16.6	18.5	15.7	9.2	-0.3	-12.7	-23.5	(33)
(34)			MCDWB	-24.1	-19.6	-5.4	4.1	14.4	24.1	25.6	21.1	12.9	1.2	-12.2	-23.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.1	8.3	12.5	12.0	9.7	10.8	10.4	10.2	8.2	6.1	6.3	5.5	(35)
(36)			MCDBR	10.4	10.6	11.8	12.1	13.6	14.5	13.8	14.1	12.9	6.3	8.8	9.3	(36)
(37)			MCWBR	10.2	10.2	9.8	8.3	7.6	6.6	5.3	6.7	7.1	5.0	8.5	9.1	(37)
(38)		5% WB	MCDWB	10.4	10.5	11.5	11.4	12.9	13.2	12.3	11.6	11.4	5.8	8.8	9.4	(38)
(39)			MCWBR	10.2	10.2	9.6	7.9	7.4	6.3	5.3	6.0	6.7	5.0	8.5	9.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.203	0.252	0.290	0.333	0.372	0.408	0.496	0.473	0.328	0.272	0.199	0.188	(40)	
(41)		taud	1.907	2.068	2.095	2.113	2.278	2.285	1.997	2.011	2.448	2.314	1.933	1.881	(41)	
(42)		Ebn at Noon	370	664	790	834	836	809	707	678	765	653	366	163	(42)	
(43)		Edh at Noon	30	65	98	121	114	116	151	136	71	51	28	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.23	1.15	2.92	5.02	5.44	5.77	5.30	3.90	2.41	0.98	0.36	0.09	(44)	
(45)		RadStd	0.01	0.05	0.09	0.17	0.32	0.39	0.30	0.32	0.26	0.11	0.03	0.00	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.88	N/A	N/A	N/A	N/A	N/A	-289	-320	+73	N/A	(46)
(47)	Regional (2 neighbors)	+0.78	N/A	N/A	N/A	N/A	N/A	-227	-277	+56	N/A	(47)

Nomenclature: See separate page